

Automatic DRIP IRRIGATION SYSTEM

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Conventional irrigation method wastes a lot of water, leading to a high cost of electricity to run the pumpset for irrigation. Automation can help save water, electricity as well as human efforts. This can be achieved with the help of a single soil-moisture sensor and an AVR microcontroller.

Here we present such an automated plant watering system that senses the moisture level of the soil and automatically switches the pump on when the power is 'on'.

Circuit and working

Circuit of the automatic drip irrigation system is shown in Fig.

transistor is used to drive relay RL1. DC motor pump M1 is connected between the normally-open (N/O) pin and pole of the relay.

If the soil is dry, i.e., the ground voltage is greater than the reference voltage, the microcontroller (ATmega16) gives a logic-1 output signal. The transistor conducts to energise

relay RL1 (as its pole pin comes in contact with the N/O pin) and the motor turns on with power supply provided from the battery.

When the soil is wet, i.e., soil voltage is less than the reference voltage, the logic-0 signal of the microcontroller turns the transistor, and hence the motor switches 'off'.

The motor pump status ('on' or 'off') and soil condition ('dry' or 'wet') are dis-

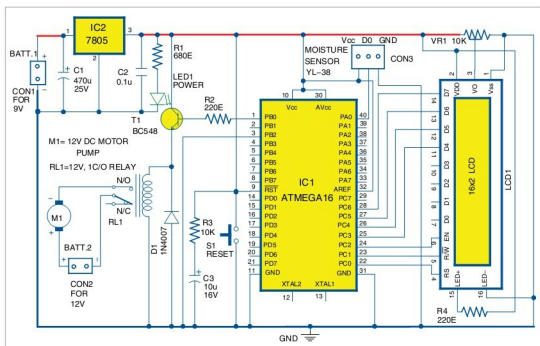


Fig. 1: Circuit of automatic drip irrigation system



Fig. 2: Author's prototype watering the plant in a flower pot

1. It has three sections: sensor, microcontroller and water-pump motor circuits.

The sensor circuit detects soil condition by measuring the soil voltage and comparing it with a reference voltage. A BC548

played on LCD1.

Fig. 2 shows the author's prototype with the moisture sensors embedded in the soil to be monitored.



Fig. 3: Author's prototype indicating the motor status in wet soil condition